EcoWatt Rule-Based Energy Analysis System

Technical Report

1. Introduction

Electricity consumption in Saudi Arabia is significantly influenced by environmental conditions, particularly high temperatures that increase the demand for cooling systems. This makes residential electricity usage highly variable across regions, seasons, and household characteristics.

The EcoWatt system is a rule-based analytical framework designed to help households understand their electricity consumption patterns by comparing their usage against a model-estimated baseline calibrated to resemble similar households, rather than a direct statistical average. The system provides:

- Historical consumption trend analysis
- Comparison with similar households
- Energy-saving advice

Unlike machine learning-based systems, EcoWatt relies on transparent, interpretable rules derived from official statistics, ensuring explainability and ease of validation.

The system is designed as a decision-support tool rather than a predictive model, and the results should be interpreted as estimations rather than exact measurements.

2. Data Sources

The system integrates multiple data sources to ensure reliability and contextual awareness:

2.1 GASTAT Household Energy Statistics (2024)

Official dataset provided by the General Authority for Statistics in Saudi Arabia [1].

It includes:

- Appliance usage patterns
- Regional segmentation
- Housing types
- Seasonal variations

This dataset is used to construct and calibrate baseline consumption profiles for households with similar characteristics through statistical aggregation and derived features rather than direct extraction.

2.2 Open-Meteo Historical Weather Data

Open-Meteo API provides historical temperature data used to [2]:

- Estimate cooling demand
- Adjust baseline consumption according to climate conditions
- Model the impact of regional temperature differences

2.3 User-Collected Survey profile

The primary dataset used in this project was collected through a structured online survey [16]. It captures real household electricity consumption along with key household characteristics, providing the foundation for analysis and validation.

The dataset includes the following attributes:

- **Monthly electricity consumption (kWh)**
  Represents the total monthly electricity usage reported by the household.
- **Monthly electricity bill cost (SAR)**
  Represents the monthly electricity bill amount corresponding to the reported consumption.

- **Region**
  Indicates the geographical location of the household across the 13 administrative regions of Saudi Arabia.
- **Housing type**
  Describes the dwelling type (e.g., apartment, villa).
- **Number of residents**
  Represents the number of people living in the household.
- **Insulation status**
  Indicates whether the household has thermal insulation (Yes/No).
- **Month**
  Represents the billing month of the recorded consumption.
- **AC type**
  Indicates the type of air conditioning system used in the household.
- **Cooking Fuel type (Electric - Gas)**
  Specifies the primary cooking energy source used by the household.

This dataset is used as the main input for analysis and validation, combining consumption values with household characteristics required for comparison with baseline models.

2.4 Energy-Saving Tips Dataset

A structured dataset of 53 electricity-saving tips was compiled from the IEA, U.S. DOE, and a ResearchGate publication.

The dataset includes the following fields:

- Type (category)
- Raw tips (original extracted content)
- Advice (processed actionable recommendation)
- Reference (source)
- Title (short heading for the advice)

LLM **Qwen2.5-3B-Instruct** was used to extract and convert the tips into clear, actionable advice.
This dataset is used to generate relevant recommendations based on the household consumption profile.

3. Data Preparation

3.1 Data Collection

In this project, the data was not taken from a single source. Instead, multiple datasets were combined to build a more realistic understanding of household electricity consumption.

The main idea is to integrate general consumption patterns with household-specific conditions in order to produce meaningful analysis.

The data sources include:

- GASTAT dataset
- Weather data (Open-Meteo)
- User survey data
- Energy-saving tips dataset

The goal of combining these sources is to connect general electricity consumption patterns with the user's actual household conditions in order to analyze energy usage in a more logical and realistic way.

3.2 Data Cleaning

Raw data is usually messy, inconsistent, and sometimes incorrect.
So before using it, we performed several cleaning steps to ensure reliability.

Key steps include:

- Text standardization (for example): "Riyadh", "riyadh", and "RIYADH" are all converted into one consistent format. This prevents matching errors later in the system.
- Handling invalid or unrealistic values (for example): Incorrect dates are automatically corrected to "2026", and bills with mismatched costs are excluded. This ensures accurate calculations.
- Handling missing values: Instead of removing data, we used median values or matched with the closest similar group.
- Categorical alignment: Ensured consistency in categories such as:
  - region
  - housing type

  - season

3.3 GASTAT Baseline Data Processing

The raw statistical data from GASTAT came in complex, multi-sheet Excel reports and scattered weather files. To make it usable for our system, we had to extract, aggregate, and structure it into a unified baseline.

The preprocessing stage includes the following steps:

- **Weather Data Aggregation:** We collected raw temperature records across different Saudi regions, calculated the monthly averages, and grouped them into two main seasons (Winter and Rest of Year).
- **Targeted Data Extraction:** Instead of using the entire massive report, we pinpointed and extracted specific energy drivers from various sheets, including:
  - HVAC usage (different AC types and heaters)
  - Appliance usage (lighting, refrigerators, and freezers)
  - Home characteristics (insulation rates and cooking fuel types)
- **Multi-Dimensional Structuring:** We combined the extracted statistics and weather data to build a comprehensive matrix. Every single row in the new dataset represents a unique combination of:
  - Region (administrative region of Saudi Arabia, e.g., Riyadh, Makkah)
  - Season (Winter and Rest of Year)
  - Housing type (Villa, Apartment, Traditional House, etc.)
- **Standardization:** All numerical values, such as weekly usage hours and percentages, were cleaned, converted into standard numeric formats, and rounded for consistency.

Result of Preprocessing:

Transform static, fragmented statistical tables into a dynamic, structured baseline (household_consumption_baseline.csv). This allows the system to instantly look up the "average Saudi household" profile to evaluate any user's consumption fairly.

3.4 Data Preprocessing

After cleaning, the dataset is transformed into a structured format suitable for analysis and comparison with the baseline dataset. This stage ensures that raw user inputs and consumption logs are converted into consistent and usable features for the system.

The preprocessing stage includes the following steps:

- User-to-baseline matching

  The user data is matched with similar households based on:
  - region
  - housing type
  - season

This allows fair comparison between households with similar characteristics.

- Data transformation:

  Raw inputs are converted into standardized analytical formats, including:
  - converting month values into seasonal categories (Winter / Rest of Year)
  - preparing numerical values (consumption, cost, etc.)
- Structuring the data for calculations so the system can compute:
  - expected consumption (baseline)
  - differences between the user and similar households

Result of Preprocessing:

A cleaned and structured user profile dataset is generated, containing standardized features and merged baseline attributes, ready for consumption analysis and energy-saving recommendation generation.

4. System Methodology

This section describes how the processed data is used to estimate consumption, calculate savings, and generate recommendations.

4.1 Appliance Power Assumptions

To estimate household energy consumption, standardized representative power values are used for common residential appliances. These values are not fixed regulatory constants issued directly by SASO, but are instead typical operating power ranges aligned with Saudi Energy Efficiency labeling frameworks and market-available appliances compliant with SASO standards [4].

Appliance	Power (kW)	Rationale & Reference
Split AC	~2.2	For a typical 24,000 BTU residential split unit, the input power under normal operating conditions is approximately 2.0–2.5 kW depending on inverter efficiency and SEER rating. The value 2.2 kW represents a realistic mid-range operating point for high-efficiency units commonly used in Saudi households under SASO energy efficiency labeling requirements.
Window AC	~3.2	Window air conditioners generally have lower efficiency compared to split systems due to integrated compressor and reduced heat exchange performance. For a 24,000 BTU unit, typical input power ranges from approximately 2.8–3.3 kW. The value 3.2 kW represents a high-consumption but realistic upper-range operating condition.
Central AC	~2.6	Ducted central AC systems introduce additional fan and duct losses. For a 24,000 BTU equivalent cooling load, typical electrical input ranges around 2.4–3.0 kW. The value 2.6 kW reflects a moderate-efficiency system under realistic residential ducted operation.
Electric Heater	~1.5	Electric resistance heaters typically operate at fixed rated loads such as 1.5 kW, 2.0 kW, or higher depending on design. The value 1.5 kW represents a common residential low-to-medium heating load used in portable heaters.
Water Heater	~2.0	Residential storage water heaters typically range between 1.5–3.0 kW depending on tank size and heating element design. The value 2.0 kW is a representative midpoint commonly used in household installations in the Saudi market.
Regular Lamp (Incandescent)	~60	60W incandescent lamps are used as a historical baseline for lighting comparisons. They are largely phased out under Saudi energy efficiency regulations, but remain a reference point for efficiency equivalence calculations.
Savings Lamp (LED)	~9	LED lamps providing equivalent luminous output to a 60W incandescent bulb typically consume around 7–10W. The value 9W represents a common mid-range efficiency level compliant with SASO lighting efficiency requirements.
Refrigerator	~150	Refrigerators do not operate at constant power; instead, compressors cycle on and off. The 150W value represents approximate compressor operating power during active cycles for a medium-sized (400–500L) modern inverter refrigerator. Actual average consumption over time is significantly lower.
Freezer	~120	Freezers operate similarly with cyclical compressor operation. The 120W value represents typical active compressor power for a standalone chest or upright freezer under normal ambient conditions, while real average consumption depends on duty cycle and insulation efficiency.

Cooking fuel type is collected but not included in consumption modeling (refer to **Assumption A.4** for details).

4.2 Baseline Consumption Model

The baseline represents an estimated expected consumption level for households with similar characteristics under typical operating conditions, rather than an exact or directly measured value (refer to **Assumption A.6**).

The model is based on the following factors:

- HVAC usage (cooling and heating separated)
- Appliance usage (water heating, lighting, refrigerators, and freezers)
- Regional temperature sensitivity
- Household characteristics (region, housing type, season)
- Occupancy scaling factor (adjusting baseline based on the number of residents)

Cooling demand is adjusted using region-specific temperature sensitivity coefficients derived from empirical research. The relationship between temperature and electricity consumption is modeled as approximately linear within practical operating ranges, consistent with findings from the KAPSARC study "*The Rising Cost of Cooling*" [8].

Weekly consumption estimates are scaled to monthly values using a calendar conversion factor (4.345 weeks per month).

Insulation effectiveness is modeled as an approximate 15% reduction in overall HVAC-related consumption (both cooling and heating), based on SEEC and KSU studies [3][9]. Unlike cooling, winter heating consumption is modeled based solely on average weekly operational hours without regional climate modifiers (refer to **Assumption A.18**).

4.3 Temperature Adjustment Model

Temperature influence is modeled using static region-specific adjustment multipliers. Rather than calculating dynamic daily temperature deviations, the system uses derived coefficients to represent the overall expected increase in cooling demand for a given climatic zone.

These values are adapted from the semi-elasticity findings in the KAPSARC study and generalized for the system as fixed regional factors:

- Central region: ~8.3%

- Western region: ~12.9%
- Southern region: ~10.3%
- Eastern region: ~4.8%

For regions not covered in the KAPSARC study (such as Qassim, Hail, Tabuk, Northern Borders, and Al-Jouf), sensitivity values are assigned by grouping them with the nearest major climatic zone (Central, Western, Southern, or Eastern).

Because the system employs a simplified region-based adjustment rather than explicit per-degree temperature deviation, the temperature modifier is applied linearly as a static markup:

$$\text{Cooling Adjustment Factor} = 1 + \text{regional sensitivity coefficient}$$

4.4 Cooling Demand Modeling

Cooling demand is mathematically modeled as a function of the following combined parameters:

- Air conditioning usage hours
- AC power ratings (kW)
- Estimated number of AC units
- AC type efficiency gap (Window, Split, Central)
- Regional cooling sensitivity multiplier
- Insulation status

AC Unit Estimation: To approximate the total cooling load, the number of active air conditioning units is estimated based on household size (refer to **Assumption A.16** for estimation rules).

Temperature & Regional Adjustments: While empirical studies (such as KAPSARC) demonstrate a linear relationship between rising temperatures and cooling demand, the baseline model simplifies this dynamic variable by applying static, region-specific sensitivity multipliers to represent overall climate impact.

AC Type Efficiency & Power Ratings: The baseline model dynamically customizes calculations based on the user's reported AC type (Split, Window, or Central). Efficiency differences are technically reflected in the system through two main variables:

1. **Standardized Power Inputs:** Each type is assigned a specific baseline power rating derived from SASO standards (e.g., Split systems are benchmarked at ~2.2 kW due to higher efficiency, while older Window units are benchmarked at ~3.2 kW due to greater thermal losses).

2. **Efficiency Gap Penalty:** The system accounts for potential optimization by applying an efficiency reduction multiplier based on international benchmarks (DOE, EnergyStar). Window units are assigned a higher potential savings gap (~8%) compared to Central systems (~3%), with Split systems treated as the optimized baseline.

4.5 Advice Selection Logic

Advice is selected based on detected consumption drivers and the overall consumption level. The selection process follows a structured logic to ensure relevance and impact:

- **Driver-to-Category Mapping:**
  The system identifies key factors contributing to high electricity consumption, such as:
  - Insulation status (mapped to "insulation" tips)
  - Air conditioning type (mapped to "cooling" tips)
  - Household occupancy (mapped to "behavior" tips)
  - Cooling demand

  Each driver is linked to a predefined category in the tips dataset to retrieve targeted recommendations.

- **Adaptive Selection:**
  The number of recommendations retrieved from each category is dynamically adjusted based on the consumption level. For instance, "High" consumption profiles receive more intensive guidance (3 tips per category) compared to "Moderate" profiles (2 tips per category).
- **Supplemental Guidance:**
  Additional recommendations from general categories (e.g., lighting or electronics) are selected based on the consumption level classification (High, Moderate, Average, Low) to provide a comprehensive energy-saving plan.

- **Refinement and Filtering:**
  To maintain user engagement and clarity:
  - **Randomized variety:** The system uses random sampling to select different tips within the same category for each analysis.
  - **Deduplication:** Overlapping recommendations are removed to ensure uniqueness.
  - **Output Constraint:** The final output is limited to a maximum of 6 prioritized advice items to avoid information overload.

The final result is a personalized set of actionable advice items strictly aligned with the user's unique consumption profile and identified inefficiencies.

4.6 Reduction (Savings) Model

The savings model estimates potential reductions based on multiple independent factors r_i:

- **Behavioral Factor (r_1):** A fixed baseline saving of 3% (0.03) is adopted for all profiles to account for sustained behavioral changes. This is consistent with IEA and DOE findings [6][7].
- **Insulation Factor (r_2):** A 15% (0.15) potential reduction, applied exclusively if the household currently lacks thermal insulation. This is based on SEEC and KSU residential energy studies.
- **AC Efficiency Factor (r_3):** Savings from optimizing cooling systems based on current AC type: 0% for Split, 3% for Central, and 8% for Window units. These values reflect the efficiency gaps identified in SASO.
- **Occupancy Factor (r_4):** To reflect the impact of household size, the system applies adjustments to both baseline consumption and potential savings for households exceeding the standard 4-resident average (refer to **Assumption A.8** for detailed calculation rates and caps).

The Multiplicative Interaction Model:

The combined reduction is calculated using a multiplicative interaction model to prevent overestimation:

$$reduction = 1 - \prod(1 - r_i)$$

Where r_i represents the contribution of each factor. These reduction factors are treated as partially independent for modeling simplicity (refer to **Assumption A.5** for a detailed rationale on factor interactions).

Maximum Reduction Cap: A maximum reduction cap of 25% is enforced as a systemic safeguard (refer to **Assumption A.7** for details).

4.7 Optimized Consumption

Optimized consumption represents a theoretical "improved scenario" where identified inefficiencies are addressed. This value is used as a performance benchmark and is not a prediction of future bills.

The system calculates this value by applying the total reduction percentage to the user’s total consumption (kWh):

$$Optimized\_kWh = Total\_kWh * (1 - Reduction)$$

This ensures the final target is personalized to the user's profile while staying within the realistic limits of the system.

4.8 Electricity Cost Model

Electricity cost is calculated using tiered pricing:

- Up to 6000 kWh (≤ 6000 kWh): 0.18 SAR/kWh
- Above 6000 kWh (> 6000 kWh): 0.30 SAR/kWh
- Value Added Tax (VAT): A 15% tax applied to the calculated consumption cost.
- Fixed Service Fee: A mandatory 15 SAR monthly meter reading fee added to the final total.

This follows the residential tariff structure defined by the Saudi Electricity Regulatory Authority [5].

5. Important Methodological Consideration

The EcoWatt model prioritizes practical interpretability over complex physical simulation. Instead of using engineering-based thermodynamic equations, the system relies on empirical relationships between regional climate conditions and electricity consumption derived from established research.

This design choice is intended to ensure:

- **Simplicity:** Keeping the logic clear and easy to implement.
- **Transparency:** Making it easy for users to understand how their results were calculated.
- **Validation:** Allowing for easier verification of results against real-world billing patterns.
- **Data Alignment:** Ensuring the model works seamlessly with available national datasets (GASTAT).

Therefore, the system should be interpreted as an approximate analytical tool designed for decision support, rather than a physically precise engineering simulation of building energy dynamics.

6. System Features

6.1 Consumption Trend Analysis

Displays up to a 6-month historical trend comparing:

- Actual consumption
- Optimized consumption

6.2 Comparison with Similar Households

The system compares the user's household with similar households selected from the baseline based on:

- Region
- Housing type
- Season

The system calculates:

- Percentage difference between actual and baseline consumption
- Consumption classification (High, Moderate, Average, Low)
- Dynamic explanatory message based on deviation

If baseline data is unavailable or fails to match, a fallback mechanism is used to ensure analytical continuity and prevent erroneous comparisons (refer to **Assumption A.9** for threshold details).

6.3 Advice Selection

By evaluating key consumption drivers such as insulation, AC type, and occupancy against the household's consumption classification, the system formulates highly tailored energy advice. To ensure the guidance remains practical, the final output is strictly capped at six prioritized tips.

7. Design Considerations

The EcoWatt system is built upon several core design principles to ensure practical utility:

- **Interpretability over Complexity:** The system follows a transparent rule-based approach rather than relying on black-box machine learning models, ensuring that every calculation can be easily explained and validated.
- **Contextual Benchmarking:** Baseline values are estimated from aggregated patterns in the GASTAT household energy dataset and adjusted using specific household characteristics. This ensures comparisons are made against "similar households" rather than a rigid national average.
- **Intentional Simplification:** Consumption estimates rely on simplified assumptions (e.g., fixed appliance power ratings) to maintain computational clarity and usability despite limited household-level granularity.
- **Variable Prioritization:** The model strategically focuses only on the most impactful variables affecting residential electricity consumption (e.g., HVAC usage, insulation, regional climate, and household occupancy) to avoid data clutter.
- **Dynamic Savings Engine:** While a minimal universal baseline saving (3%) is assumed for positive behavioral changes, the primary reduction targets are derived dynamically from multiple household-specific inefficiencies. To prevent compounded overestimation, these combined reductions are strictly limited by a 25% system-level cap.

8. Assumptions

The system relies on several assumptions to simplify modeling and maintain interpretability:

- A.1 Appliance usage is approximated using fixed average power ratings rather than exact device specifications.
- A.2 Temperature influence on consumption is modeled using heuristic adjustment factors instead of physical cooling load equations.
- A.3 Household behavior is approximated as consistent within groups defined by region, season, and housing type.
- A.4 Cooking energy consumption is excluded due to variability and lack of reliable measurement data, which may slightly underestimate total household electricity consumption.
- A.5 Energy savings are modeled using relative multiplicative effects rather than fixed additive reductions. While some interaction between factors may exist in reality (e.g., insulation and cooling demand), the multiplicative formulation provides a balanced approximation without overestimating combined effects.
- A.6 Baseline consumption is treated as a model-estimated reference rather than an exact expected value for any individual household.
- A.7 A maximum reduction cap of 25% is enforced as a systemic safeguard. This prevents the multiplicative model from mathematically overestimating total savings when combining multiple efficiency factors, ensuring the outputs remain realistic.
- A.8 Occupancy effects are modeled using bounded heuristic mechanisms to prevent extreme scaling:
  - a. An approximate 1.5% increase in baseline consumption per additional resident (strictly capped between -5% and +15% of the total multiplier).
  - b. A 2% contribution in savings estimation (capped at a maximum of 5%).
- A.9 Baseline comparison calculations include a numerical stability adjustment where extremely small baseline values are constrained to a minimum safety threshold (200 kWh) to prevent distortion in percentage-based comparisons.
- A.10 Energy consumption is approximated assuming constant average power usage during operating hours, without explicitly modeling variable load behavior (e.g., inverter-based AC cycling), due to data limitations.
- A.11 Thresholds used for driver detection (e.g., >20% for High consumption) are heuristic and chosen for interpretability rather than strict statistical derivation.

A.12 Appliance usage hours are assumed to follow GASTAT averages. Differences between actual and baseline consumption are attributed to efficiency factors rather than usage duration, which may misclassify households with atypical usage patterns.

A.13 AC type affects both power consumption and selected usage hours, while the usage patterns themselves are based on aggregated GASTAT averages.

A.14 Water heater thermodynamics are not fully modeled. While usage hours vary seasonally based on GASTAT data, the physical energy required to heat colder winter water is ignored, relying instead on a constant 2.0 kW power assumption.

A.15 Key variables such as dwelling size (square meters) were not included due to data unavailability. These factors directly influence HVAC loads and their exclusion may introduce estimation bias.

A.16 The total number of active air conditioning units per household is heuristically estimated based on occupancy, assuming approximately one operating AC unit per two residents (with a strictly enforced minimum of one unit). This simplification avoids requiring complex user inputs regarding simultaneous appliance usage.

A.17 In cases where regional statistical baseline data lacks specific thermal insulation adoption rates, the system programmatically assumes a 50% regional insulation rate (a 0.5 fallback value) to ensure computational continuity.

A.18 Winter heating consumption is modeled based solely on average operational hours without applying regional climate sensitivity multipliers. This assumes that heating loads are relatively minor and uniform across the limited Saudi winter season compared to the highly variable summer cooling loads.

These assumptions are necessary to balance simplicity, interpretability, and practical usability of the system.

9. Future Work

The current EcoWatt system is built on a rule-based approach for energy analysis. In future updates, the goal is to improve accuracy and make the recommendation system more advanced.

- **Dynamic Climate Modeling:** Instead of using fixed regional temperature factors, the system will move toward a more dynamic model. Based on insights from KAPSARC, linking monthly temperature changes directly to cooling demand using a linear relationship can improve the accuracy of energy estimation and make the system more responsive to real weather variations.

- **LLM Integration:** At the moment, the system relies on predefined rules to generate recommendations. In future versions, a Large Language Model (LLM) will be introduced to enhance this process. The LLM will help generate more personalized energy-saving advice based on user-specific household conditions, while still staying consistent with the outputs of the rule-based engine. This personalization will also extend to the recommendation titles, ensuring they are tailored to each user. It will mainly improve how the recommendations are delivered, making them more natural and easier to understand for users.

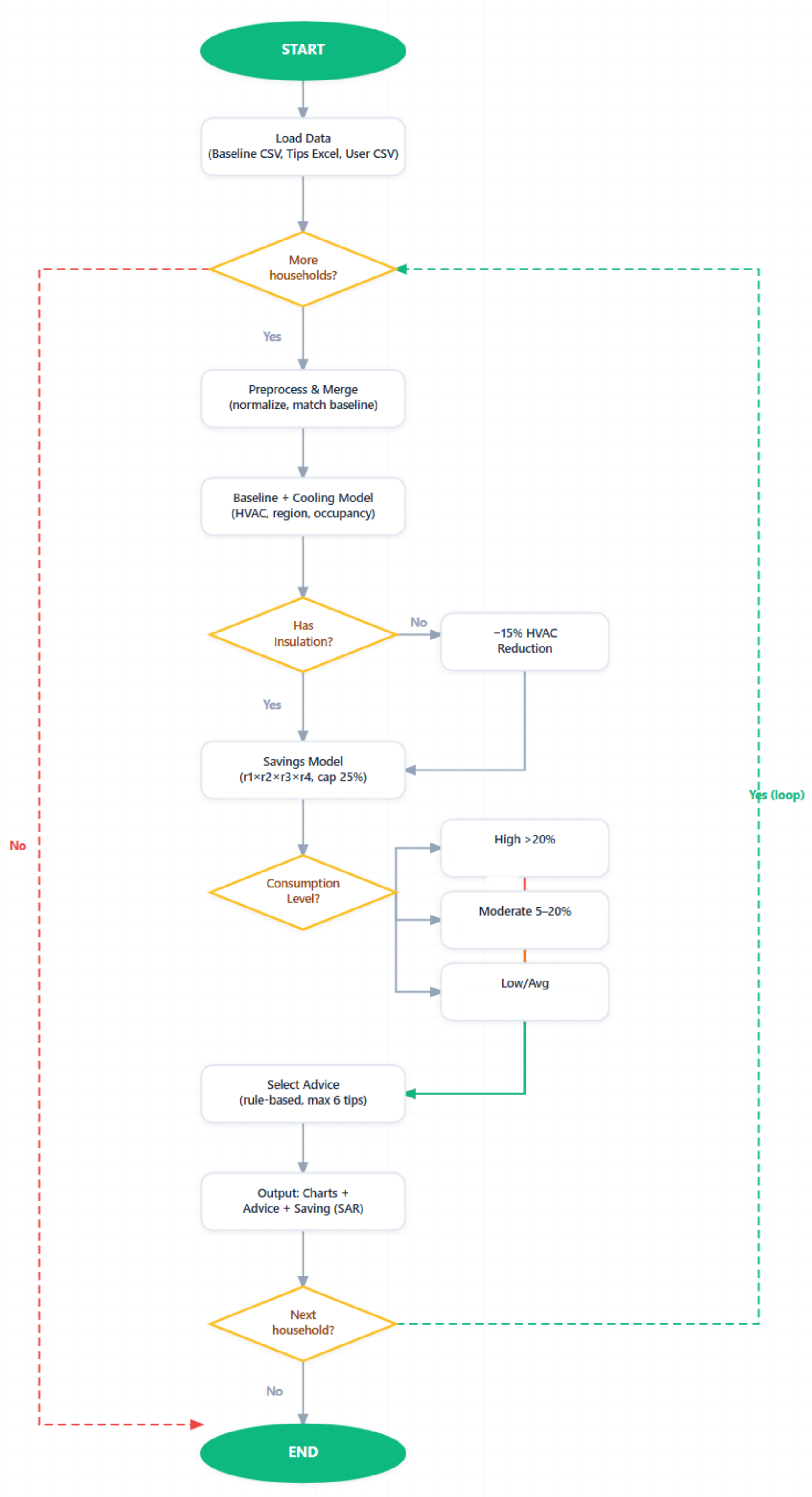
10. Conclusion

The EcoWatt project presents a practical rule-based approach for analyzing household electricity consumption by integrating official statistical data with specific household characteristics. The system is designed as a decision-support tool for the residential sector in Saudi Arabia, allowing users to compare their consumption against model-based regional baselines.

By identifying the main drivers of electricity usage and providing targeted recommendations, EcoWatt offers a structured way to evaluate household energy efficiency. While the system is based on simplified assumptions and aggregated statistical patterns, it provides a useful analytical framework for understanding consumption behavior and supporting basic energy-saving decisions.

11. System Execution Flowchart

This flowchart illustrates the EcoWatt rule-based pipeline, from data input and baseline matching to generating personalized energy-saving advice.



12.References

[1] General Authority for Statistics (GASTAT). *Household Energy Statistics Reports*. Available: https://www.stats.gov.sa/en/statistics-tabs/-/categories/124707?tab=436312&category=124707.

[2] *Open-Meteo*, Historical Weather API. [Online]. Available: https://open-meteo.com

[3] Saudi Energy Efficiency Center (SEEC). *Energy Efficiency Guidelines and Reports*. Available: https://www.seec.gov.sa

[4] Saudi Standards, Metrology and Quality Organization (SASO). *SASO 2663:2018 – Minimum Energy Performance, Labelling and Testing Requirements for Air Conditioners*. Available: https://www.saso.gov.sa

[5] Saudi Electricity Regulatory Authority (SERA). *Electricity Tariff – Residential Consumption*. Available: https://www.sera.gov.sa/en/consumer/electric-tariff/electric-tariff-categories/consumption-tariff

[6] International Energy Agency (IEA). *Energy Efficiency Indicators and Residential Energy Use*. Available: https://www.iea.org

[7] U.S. Department of Energy. *Appliance and Electronics Energy Use*. Available: https://www.energy.gov/energysaver

[8] King Abdullah Petroleum Studies and Research Center (KAPSARC). *The Rising Cost of Cooling*. Available: https://www.kapsarc.org/our-offerings/publications/the-rising-cost-of-cooling-regional-energy-futures-in-a-warming-saudi-arabia/

[9] King Saud University (KSU), *Resalat Al-Jami'ah Magazine*. Thermal Insulation and Its Impact on Electricity Consumption. Available: https://rs.ksu.edu.sa/ar/issue-1247/136

[10] Hassan H. Umar & Omar S. Asfour (2025). Retrofit strategies to improve energy efficiency through the integration of thermal insulation into the residential buildings of Saudi Arabia. *Case Studies in Thermal Engineering,* 73, 106620. ScienceDirect. Available: https://doi.org/10.1016/j.csite.2025.106620

[11] U.S. Department of Energy (DOE). *Air Conditioning Energy Saver Guide*. Available: https://www.energy.gov/energysaver/air-conditioning

[12] Energy Star. *Most Efficient Residential Cooling Systems*. Available: https://www.energystar.gov/products/most_efficient

[13] A. M. Al-Ghaili, et al., "Residential air conditioning energy consumption studies: A state-of-the-art review," *Journal of Building Engineering*, vol. 51, p. 104332, 2022. https://www.sciencedirect.com/science/article/abs/pii/S235271022200657X

[14] MDPI. *Evaluating and Enhancing the Energy Efficiency of Residential Buildings*. Available: https://www.mdpi.com/1996-1073/15/20/7763

[15] National Center for Biotechnology Information (NCBI). *The Climate and Health Benefits from Intensive Building Energy Efficiency Improvements*. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC8378816/

[16] Google Forms, *"Household Electricity Consumption Survey,"* 2026. [Online]. Available: https://docs.google.com/forms/d/1N4cQmG4KnAinCYwpkX0ydJl2cr5ye09Y2oOaFZPV-Sk/viewform